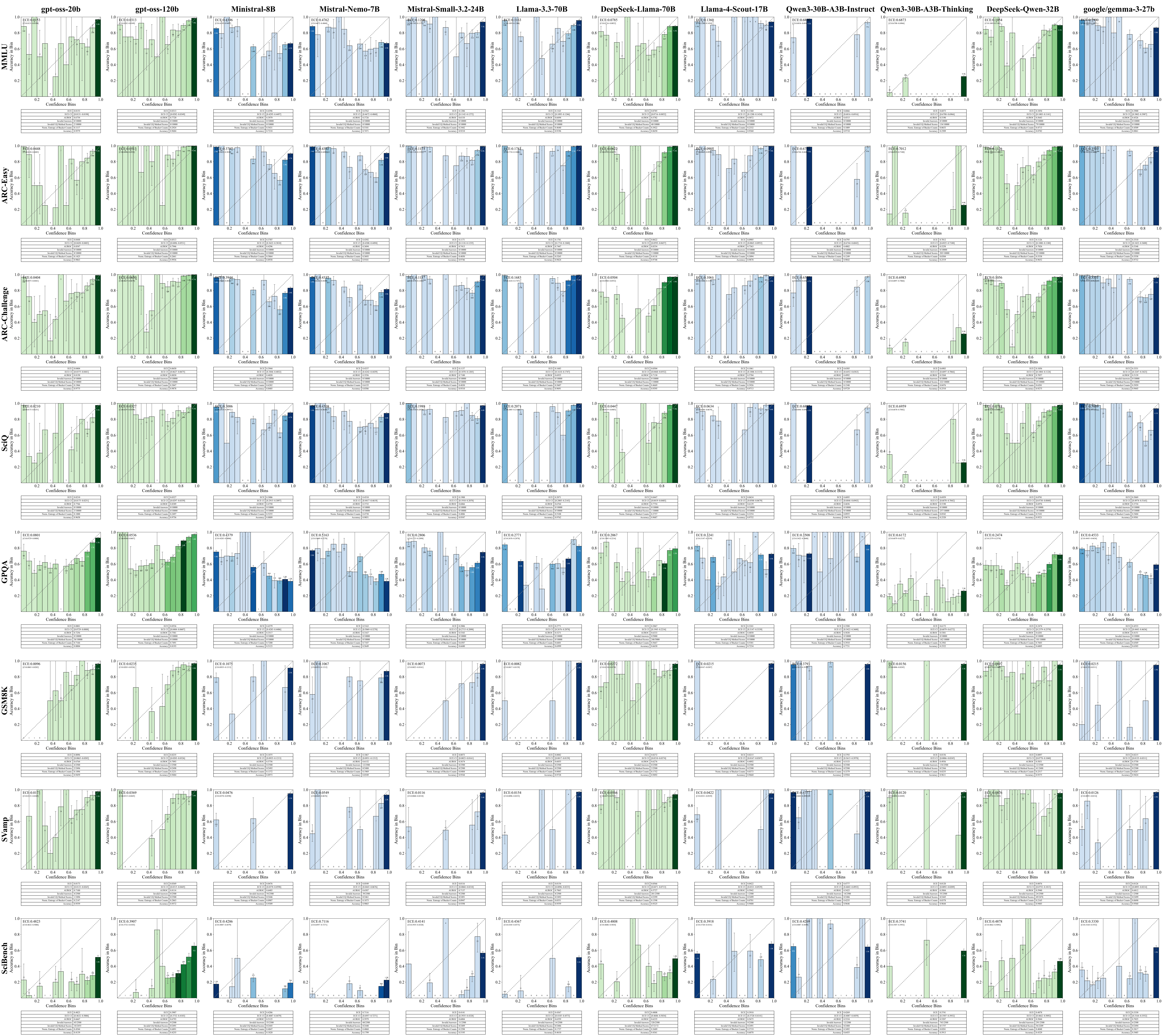
Calibration Plots for Uncertainty Method Verbalized Uncertainty



MMLU

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	DeepSeek-Llama-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0153	0.0313	0.4396	0.4762	0.1206	0.1163	0.0785	0.1360	0.6866	0.6873	0.1054	0.3900
ECE CI	[0.0131–0.0198]	[0.0282–0.0349]	[0.4295–0.4497]	[0.4672–0.4868]	[0.1142–0.1272]	[0.1095–0.1246]	[0.0716–0.0852]	[0.1288–0.1434]	[0.6816–0.6916]	[0.6788–0.6966]	[0.1003–0.1141]	[0.3802–0.3987]
AUROC	0.6754	0.7720	0.3979	0.3726	0.6110	0.6694	0.5782	0.5072	0.4515	0.5190	0.5643	0.4526
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	2/10000	0/10000	0/10000	0/10000	0/10000	0/10000	40/10000	0/10000	0/10000	134/10000	79/10000	0/10000
Norm. Entropy of Bucket Counts	0.1117	0.2924	0.5431	0.5181	0.3442	0.4640	0.3922	0.2332	0.1089	0.0653	0.4713	0.4509
Accuracy	0.9579	0.9684	0.7218	0.7471	0.9210	0.9196	0.8650	0.9360	0.9710	0.2509	0.8782	0.8861

ARC-Easy

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	DeepSeek-Llama-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0448	0.0513	0.3707	0.4392	0.1171	0.1781	0.0622	0.0905	0.6795	0.7012	0.1126	0.3510
ECE CI	[0.0420–0.0483]	[0.0496–0.0531]	[0.3622–0.3810]	[0.4308–0.4494]	[0.1118–0.1235]	[0.1718–0.1840]	[0.0593–0.0657]	[0.0865–0.0952]	[0.6744–0.6842]	[0.6925–0.7100]	[0.1080–0.1180]	[0.3422–0.3600]
AUROC	0.8547	0.7946	0.4296	0.3694	0.7413	0.7447	0.8354	0.7363	0.4882	0.5258	0.7826	0.5348
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	8/10000	0/10000	0/10000	0/10000	0/10000	0/10000	3/10000	0/10000	0/10000	203/10000	22/10000	0/10000
Norm. Entropy of Bucket Counts	0.1425	0.2661	0.5864	0.5693	0.4050	0.5265	0.4118	0.3094	0.1249	0.0384	0.5538	0.5558
Accuracy	0.9863	0.9936	0.8938	0.9181	0.9746	0.9824	0.9748	0.9878	0.9842	0.2519	0.9652	0.9732

ARC-Challenge

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	DeepSeek-Llama-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0404	0.0650	0.3944	0.4337	0.1137	0.1683	0.0504	0.1061	0.6505	0.6983	0.1056	0.3331
ECE CI	[0.0375–0.0441]	[0.0627–0.0675]	[0.3846–0.4032]	[0.4242–0.4439]	[0.1078–0.1203]	[0.1618–0.1747]	[0.0460–0.0552]	[0.1006–0.1115]	[0.6452–0.6563]	[0.6897–0.7066]	[0.1004–0.1124]	[0.3247–0.3423]
AUROC	0.8150	0.8830	0.4020	0.3556	0.7100	0.6935	0.7130	0.5704	0.4992	0.5268	0.7057	0.5139
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	3/10000	0/10000	0/10000	222/10000	35/10000	0/10000
Norm. Entropy of Bucket Counts	0.1966	0.3847	0.6282	0.6155	0.4830	0.5607	0.4669	0.3926	0.1548	0.0476	0.6201	0.5959
Accuracy	0.9775	0.9878	0.8405	0.8586	0.9510	0.9691	0.9395	0.9713	0.9729	0.2510	0.9275	0.9445

SciQ

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	DeepSeek-Llama-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0210	0.0327	0.3006	0.4510	0.1988	0.2071	0.0447	0.0634	0.6895	0.6959	0.0781	0.5069
ECE CI	[0.0175–0.0251]	[0.0297–0.0359]	[0.2915–0.3097]	[0.4417–0.4619]	[0.1910–0.2070]	[0.2005–0.2143]	[0.0419–0.0485]	[0.0588–0.0679]	[0.6846–0.6943]	[0.6870–0.7042]	[0.0730–0.0840]	[0.4974–0.5165]
AUROC	0.7766	0.8389	0.4750	0.3732	0.6066	0.6764	0.7754	0.6129	0.4656	0.5190	0.6976	0.3734
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	3/10000	0/10000	0/10000	0/10000	0/10000	0/10000	1/10000	0/10000	0/10000	257/10000	14/10000	0/10000
Norm. Entropy of Bucket Counts	0.1000	0.2497	0.5255	0.4738	0.4086	0.5108	0.3757	0.2254	0.0955	0.0542	0.5101	0.5404
Accuracy	0.9658	0.9756	0.8809	0.9051	0.9547	0.9762	0.9687	0.9752	0.9674	0.2520	0.9521	0.9501

GPQA

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	DeepSeek-Llama-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0801	0.0536	0.4379	0.5163	0.2806	0.2771	0.2067	0.3241	0.3508	0.6172	0.2474	0.4533
ECE CI	[0.0729–0.0888]	[0.0484–0.0607]	[0.4283–0.4486]	[0.5069–0.5270]	[0.2715–0.2898]	[0.2676–0.2870]	[0.1945–0.2216]	[0.3147–0.3339]	[0.3422–0.3600]	[0.6079–0.6272]	[0.2379–0.2570]	[0.4443–0.4636]
AUROC	0.7256	0.7581	0.3517	0.3167	0.5343	0.5271	0.6333	0.4850	0.5830	0.5301	0.6201	0.4151
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/5000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	36/10000	0/10000	0/10000	0/10000	0/10000	0/10000	248/5000	0/10000	0/10000	142/10000	101/10000	0/10000
Norm. Entropy of Bucket Counts	0.7166	0.7491	0.7200	0.7187	0.6587	0.6601	0.5667	0.5301	0.3910	0.2502	0.7603	0.6569
Accuracy	0.8004	0.8193	0.5123	0.5649	0.6680	0.6999	0.6630	0.7234	0.7731	0.2522	0.6005	0.6303

GSM8K

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	DeepSeek-Llama-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0096	0.0235	0.1075	0.1067	0.0073	0.0082	0.0272	0.0215	0.3793	0.0156	0.0897	0.0215
ECE CI	[0.0065–0.0202]	[0.0203–0.0326]	[0.0935–0.1213]	[0.0931–0.1212]	[0.0023–0.0161]	[0.0017–0.0139]	[0.0210–0.0374]	[0.0167–0.0307]	[0.3615–0.3978]	[0.0086–0.0245]	[0.0778–0.1040]	[0.0135–0.0311]
AUROC	0.6764	0.7093	0.5756	0.6152	0.5674	0.6055	0.6276	0.4892	0.5153	0.4936	0.5831	0.5750
Invalid Answers	4/2500	2/2500	8/2500	12/2500	0/2500	0/2500	0/2500	0/2500	0/2500	134/2500	12/2500	0/2500
Invalid UQ Method Scores	2/2496	0/2498	0/2492	0/2488	0/2500	0/2500	2/2500	0/2500	0/2500	0/2366	41/2488	0/2500
Norm. Entropy of Bucket Counts	0.1896	0.3231	0.1372	0.1969	0.0484	0.0085	0.1752	0.0373	0.3379	0.0255	0.2517	0.0267
Accuracy	0.9459	0.9484	0.8973	0.8549	0.9544	0.9728	0.9492	0.9660	0.9584	0.9611	0.9375	0.9424

SVamp

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	DeepSeek-Llama-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0173	0.0369	0.0476	0.0549	0.0116	0.0154	0.0566	0.0422	0.4777	0.0120	0.0876	0.0126
ECE CI	[0.0123–0.0265]	[0.0315–0.0445]	[0.0370–0.0598]	[0.0443–0.0676]	[0.0068–0.0210]	[0.0096–0.0233]	[0.0471–0.0713]	[0.0321–0.0529]	[0.4602–0.4952]	[0.0092–0.0209]	[0.0752–0.1013]	[0.0093–0.0214]
AUROC	0.7100	0.8114	0.6485	0.6587	0.7137	0.7065	0.5727	0.5942	0.5425	0.5884	0.5940	0.6812
Invalid Answers	4/2500	0/2500	54/2500	39/2500	9/2500	8/2500	201/2500	1/2500	14/2500	55/2500	24/2500	1/2500
Invalid UQ Method Scores	1/2496	0/2500	0/2446	0/2461	0/2491	0/2492	4/2299	0/2499	0/2486	0/2445	49/2476	0/2499
Norm. Entropy of Bucket Counts	0.2147	0.2863	0.0907	0.1873	0.0947	0.0551	0.1598	0.0701	0.4255	0.0378	0.2165	0.0690
Accuracy	0.9559	0.9572	0.9309	0.9065	0.9398	0.9559	0.9325	0.9400	0.9646	0.9644	0.9485	0.9560

SciBench

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	DeepSeek-Llama-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.4823	0.3907	0.4286	0.7116	0.4141	0.4367	0.4808	0.3918	0.4269	0.3741	0.4878	0.3330
ECE CI	[0.4632–0.5006]	[0.3732–0.4103]	[0.4087–0.4479]	[0.6957–0.7271]	[0.3935–0.4328]	[0.4185–0.4573]	[0.4606–0.5034]	[0.3720–0.4141]	[0.4087–0.4459]	[0.3507–0.3952]	[0.4662–0.5092]	[0.3164–0.3552]
AUROC	0.6667	0.6793	0.5125	0.5979	0.6084	0.6559	0.6235	0.5675	0.5334	0.5287	0.5844	0.7025
Invalid Answers	45/2500	9/2500	55/2500	60/2500	11/2500	14/2500	328/2500	9/2500	12/2500	743/2500	291/2500	6/2500
Invalid UQ Method Scores	10/2455	0/2491	0/2445	0/2440	0/2489	0/2486	67/2172	0/2491	0/2488	0/1757	56/2209	0/2494
Norm. Entropy of Bucket Counts	0.4544	0.6891	0.4847	0.4068	0.2298	0.1737	0.3689	0.3719	0.3492	0.0281	0.4048	0.3817
Accuracy	0.4139	0.4255	0.1820	0.1799	0.5195	0.4610	0.4361	0.6078	0.6455	0.5913	0.4329	0.5485